NGUYỄN MINH QUÂN

SOFTWARE ENGINEER

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SUMMARY

Software Engineer with 2+ years building scalable, high-performance systems across construction-tech (BIM/IFC), enterprise and e-learning platforms. Strong foundation in system design, clean architecture, multi-tenant SaaS and event-driven processing for high-concurrency workloads. Known for full-cycle ownership — from database and backend architecture to CI/CD, code-quality infrastructure and security — and for using AI coding assistants (Claude, Codex) to ship faster without compromising quality.

EDUCATION

January 2025

Software Engineering | FPT College
GPA: 3.7 / 4.0

Achievements:

- Top 100 Students – Spring 2024 Semester
- Top 20 – FE Hackathon 2024
- Graduated with Distinction from Cybersoft Academy

SKILLS

Languages: TypeScript, C#, PHP

Frameworks & Libraries: NestJS, .NET, Laravel, Next.js, Angular

API & Architecture: RESTful API, GraphQL, Multi-tenant Architecture (tenant isolation, per-tenant configuration), Event-driven Architecture, Micro-Frontend, SOLID, Design Patterns, Scalable System Design

Database, Cache & Messaging: PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ

DevOps & Tools: Git, Docker, GitLab CI/CD, Nginx, AWS, Linux, OPNsense

Code Quality Infrastructure: SonarQube (self-hosted setup), reusable GitLab CI templates that auto-scan every merge request across frameworks, Quality Gates

AI-Assisted Development: Claude (Claude Code), OpenAI Codex — code generation, refactoring, code review, test writing and technical documentation

EXPERIENCES

TGL Solutions

(December 2024 – Present)

> **TGL Spot Award – Q3/2026:** Recognized as the top-performing employee of the quarter.

Fullstack Developer | PMS – Project Management System

Description: A construction project management platform (similar to Microsoft Project) for managing projects, materials and labor, with approval workflows for project managers.

Key features:

- AI-powered task estimation from 3D IFC models using Google Gemini
- Self-improving estimation: historical customer project data is fed back to Gemini as context (RAG), increasing accuracy over time
- Task-level acceptance (inspection & sign-off) workflow for each project task
- Multi-level approval workflows for project managers
- Material and labor management linked to project tasks and schedules

Tech stack: Laravel, Next.js, PostgreSQL, Docker, Nginx, Google Gemini, RAG

Backend Developer | AiCest – Steel Structure Estimation from IFC *(in parallel with PMS)*
Description: A quotation system for steel structure projects that performs quantity takeoff directly from 3D IFC models and generates cost quotations.
Key features:

- Quantity takeoff and analysis of 1,000,000+ elements in a single IFC file
- Low-latency, high-performance takeoff and quotation processing
- Streaming upload for very large IFC files
- Asynchronous processing pipeline with message queue and caching

Tech stack: C# (.NET), RabbitMQ, PostgreSQL, Redis

Backend Developer | WMS – Work Management Module
Description: A multi-tenant work management module shared across multiple products (including PMS), built to handle large-scale projects with 10,000+ tasks each.
Key features:

- Multi-tenant architecture — one module reused by many projects and products
- Kanban & List views with drag-and-drop reordering powered by the LexoRank algorithm
- Import/Export of MS Project (.mpp) and Excel files
- AI-generated construction Gantt charts from user prompts
- Modular, multi-platform design using Factory & Strategy patterns

Tech stack: NestJS, PostgreSQL, Docker, Generative AI

Fullstack Developer | NewCAD Web – Integrated Enterprise Suite
Description: An integrated enterprise suite (CRM, SFA, EMS, LMS, IMS), refactored and modernized from a legacy system.
Key features:

- Refactored multi-module system with improved maintainability and faster response times than the legacy version
- Application protection with a Web Application Firewall (WAF)
- Network security and server environment management with OPNsense

Tech stack: Laravel, Next.js, WAF, OPNsense

My Vo Technology and Media

(October 2024 – Present)

Fullstack Developer & Mentor | LMS – E-Learning Platform (Remote)
Description: An online learning platform where instructors upload, store and manage lectures and students watch lessons — built and operated end-to-end, plus mentoring students in web development.
Key features:

- Video upload and streaming via Bunny.net (Storage & Stream)
- Role-based user permissions for instructors and students
- Content protection to prevent video source leaking
- Automated CI/CD deployment to VPS

Tech stack: NestJS, Next.js, PostgreSQL, Bunny.net, CI/CD, VPS